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Information processing apparatus, information processing method, and non-transitory computer readable medium for assignment of a parcel as a parking slot

Abstract

An information processing apparatus includes an interface configured to be able to acquire information regarding a parcel, and a controller configured to generate, based on the information regarding the parcel, information regarding assignment of the parcel as a parking slot.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5646853	12/1996	Takahashi	348/149	G08G 1/08
2015/0088790	12/2014	Chidlovskii	706/12	G06Q 30/0202
2017/0267233	12/2016	Minster	N/A	B60W 30/06
2018/0253101	12/2017	Banzhaf	N/A	G05D 1/0212
2020/0168091	12/2019	Batra	N/A	G08G 1/065

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2000-76423	12/1999	JP	N/A
2002-74431	12/2001	JP	N/A
2002-169957	12/2001	JP	N/A
2016-19166	12/2015	JP	N/A
2019-120982	12/2018	JP	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-142479, filed on Sep. 7, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an information processing apparatus, an information processing method, and a non-transitory computer readable medium.

BACKGROUND

(3) Parking space rental and management systems to smoothly match demands for parking spaces with parking spaces in underutilized conditions by disclosing real-time usage status of the parking spaces are known (see, for example, Patent Literature (PTL) 1).

CITATION LIST

Patent Literature

(4) PTL 1: JP 2019-120982 A

SUMMARY

(5) Not only spaces available as parking slots but also vacant spaces that can be used for parking are required to be used as parking spaces.

(6) It would be helpful to make effective use of vacant spaces that can be used for parking, as parking spaces.

- (7) An information processing apparatus according to an embodiment of the present disclosure includes: an interface configured to be able to acquire information regarding a parcel; and a controller configured to generate, based on the information regarding the parcel, information regarding assignment of the parcel as a parking slot.
- (8) An information processing method according to an embodiment of the present disclosure includes: acquiring information regarding a parcel; and generating, based on the information regarding the parcel, information regarding assignment of the parcel as a parking slot.
- (9) A non-transitory computer readable medium according to an embodiment of the present disclosure stores an information processing program. The information processing program is configured to cause a processor to execute operations, the operations including: acquiring information regarding a parcel; and generating, based on the information regarding the parcel, information regarding assignment of the parcel as a parking slot.
- (10) According to the information processing apparatus, the information processing method, and the non-transitory computer readable medium of the embodiments of the present disclosure, vacant spaces that can be used for parking can be effectively used as parking spaces.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In the accompanying drawings:
- (2) FIG. 1 is a block diagram illustrating an example configuration of an information processing system according to an embodiment;
- (3) FIG. 2 is a schematic diagram illustrating an example of a parcel to be assigned as a parking slot;
- (4) FIG. 3 is a flowchart illustrating an example procedure for an information processing method according to the embodiment; and
- (5) FIG. 4 is a flowchart illustrating an example procedure for the information processing method according to another embodiment.

DETAILED DESCRIPTION

(6) (Configuration of Information Processing System 1)

(7) An information processing system 1 according to an embodiment provides a service that finds parcels that can be used as parking slots from among various vacant spaces in business districts, residential areas, or the like, and suggests the use of the parcels as parking slots to owners of the parcels. The information processing system 1 may provide a service that actually assigns and manages the parcels as parking slots. The information processing system 1 may provide a service that introduces the parcels that can be used as parking slots to users who are looking for parking slots. The information processing system 1 may provide a service that links a service that suggests the use of parcels as parking slots with a service that manages the parking slots or introduces the parking slots. Owners of parcels to be used as parking slots, or contractors entrusted by the owners to manage the parcels are also referred to as lending users. Users who use the parking slots are also referred to as using users.

(8) As illustrated in FIG. 1, the information processing system 1 according to an embodiment includes an information processing apparatus 10 and an information acquisition apparatus 40. The information acquisition apparatus 40 acquires information regarding parcels. The information regarding parcels is also referred to as parcel information. The parcel information may include information on vacant spaces in a business district, residential area, or the like. The parcel information may include information on spaces that are temporarily vacant. As illustrated in FIG. 2, for example, the parcel information may include information regarding a first parcel 110 or a second parcel 120 that is present in front of a house 100.

(9) The information processing apparatus **10** may determine, based on the parcel information, whether the parcel can be used as a parking slot. The information processing apparatus **10** may extract, based on the information regarding parcels including various vacant spaces, a parcel that can be used as a parking slot from the vacant spaces. In the example of FIG. 2, the information processing apparatus **10** may extract the vacant first parcel **110**, as a parcel that can be used as a parking slot. The information processing apparatus **10** may extract the second parcel **120** in which a vehicle **130** is parked, as a parking slot that can be used while the vehicle **130** is out of the garage.

(10) An example configuration of the information processing system **1** will be described below in the concrete.

(11) <Information Processing Apparatus **10**>

(12) The information processing apparatus **10** includes a controller **12**, a memory **14**, and an interface **16**.

(13) The controller **12** controls at least one component of the information processing apparatus **10**. The controller **12** may be configured to include at least one processor. The “processor” is a general purpose processor, a dedicated processor specialized for specific processing, or the like in the present embodiment but not limited to these. The controller **12** may be configured to include at least one dedicated circuit. The dedicated circuit may include, for example, a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC). The controller **12** may be configured with the dedicated circuit instead of the processor, or may be configured with the dedicated circuit along with the processor.

(14) The memory **14** may be configured to include a semiconductor memory, a magnetic memory, an optical memory, or the like, for example, but not limited to these. The memory **14** may function as, for example, a main memory, an auxiliary memory, or a cache memory. The memory **14** may be configured to include an electromagnetic storage medium, such as a magnetic disk. The memory **14** may be configured to include a non-transitory computer readable medium. The memory **14** stores any information or program to be used for operations of the information processing apparatus **10**. For example, the memory **14** may store a system program, an application program, or the like. The memory **14** may be included in the controller **12**.

(15) Information, data, or the like is outputted from and inputted to the controller **12** through the interface **16**. The interface **16** may include a communication module for communication with other apparatuses, such as the information acquisition apparatus **40**, via a network **30**. The communication module may be, for example, compliant with a mobile communication standard, such as the 4th Generation (4G) standard or the 5th Generation (5G) standard. The communication module may be compliant with a communication standard, such as a Local Area Network (LAN). The communication module may be compliant with a wired or wireless communication standard. The communication module is not limited to these examples and may be compliant with various communication standards. The interface **16** may be configured to be able to connect to a communication module.

(16) The interface **16** may be configured with an input device for accepting input of information, data, or the like from a user. The input device may be configured with, for example, a touch panel, a touch sensor, or a pointing device such as a mouse. The input device may be configured with a physical key. The input device may be configured with an audio input device, such as a microphone. The interface **16** may be configured to be able to connect to an external input device. The interface **16** may be configured to be able to acquire, from the external input device, information or data input to the external input device.

(17) The interface **16** may be configured to include an output device that outputs information, data, or the like to the user. The output device may include, for example, a display device that outputs visual information, such as images, letters, or graphics. The display device may be configured with, for example, a Liquid Crystal Display (LCD), an organic or inorganic Electro-Luminescence (EL) display, a Plasma Display Panel (PDP), or the like. The display device is not limited to the above

displays and may be configured with various other types of displays. The display device may be configured with a light emitting device, such as a Light Emitting Diode (LED) or a Laser Diode (LD). The display device may be configured with various other devices. The output device may include, for example, an audio output device, such as a speaker that outputs audio information, such as voice. The output device is not limited to the above examples and may include various other devices. The interface **16** may be configured to be able to connect to an external output device. The interface **16** may be configured to be able to output information or data to the external output device.

(18) The information processing apparatus **10** may be configured to include a single server apparatus or a plurality of server apparatuses capable of communicating with each other. The information processing apparatus **10** may be realized as a cloud server.

(19) <Information Acquisition Apparatus **40**>

(20) The information acquisition apparatus **40** acquires parcel information and outputs the parcel information to the information processing apparatus **10**.

(21) The parcel information may include information representing a track record of a vacant status of a parcel. The information representing a track record of a vacant status of a parcel may include a time slot in which the parcel is unoccupied without the presence of a vehicle or an object such as luggage, or the length of time for which the parcel is unoccupied. When multiple parcels can be set up in a space, the information representing a track record of a vacant status of a parcel may include the number of vacant parcels in the space. The information representing a track record of a vacant status of a parcel may include information representing the number or size of vehicles that can be parked when the parcel is vacant. The information acquisition apparatus **40** may acquire the information representing a track record of a vacant status of a parcel, as images of the parcel. The information acquisition apparatus **40** may acquire the information representing a track record of a vacant status of a parcel, as detection results from a sensor installed in the parcel.

(22) The information acquisition apparatus **40** may be configured as an in-vehicle camera mounted on a vehicle traveling on roads around a certain parcel, for example, to acquire images of the parcel as the parcel information. The information acquisition apparatus **40** may be configured as an in-vehicle camera mounted on a route bus or the like that regularly travels on roads around a certain parcel. The information acquisition apparatus **40** may output the images of the certain parcel captured by the in-vehicle camera, as information regarding the certain parcel. The information acquisition apparatus **40** may be configured, for example, as a movable or fixed surveillance camera that is installed to include a certain parcel in an imaging range. The information acquisition apparatus **40** may output images of the certain parcel captured by the surveillance camera, as information regarding the certain parcel. The information acquisition apparatus **40** may be configured, for example, as a drone, helicopter, aircraft, satellite, or the like that takes aerial photographs of an area including a certain parcel. The information acquisition apparatus **40** may output the aerial photographs of the certain parcel, as information regarding the certain parcel.

(23) The information acquisition apparatus **40** may output, as information regarding the certain parcel, information associating the dates and times or location information at which the certain parcel is imaged with the images of the certain parcel. The information acquisition apparatus **40** may output, as information regarding the certain parcel, a combination of at least two images of the same parcel at at least two different dates and/or times. The information acquisition apparatus **40** may generate information representing a track record of a vacant status of the certain parcel based on the images of the certain parcel.

(24) The parcel information may include the amount of traffic on roads passing through a predetermined area including a parcel. The information acquisition apparatus **40** may, for example, include a camera that images roads around a certain parcel and calculate the amount of traffic based on images of the roads around the certain parcel. The information acquisition apparatus **40** may include an interface to acquire the amount of traffic on roads around the parcel from an external

apparatus such as a traffic control server. The information acquisition apparatus **40** may include an external apparatus such as a traffic control server. The parcel information may include, as the amount of traffic on roads around a parcel, the amount of traffic during morning or evening commuting hours, during daytime or nighttime hours, or the like. The parcel information may include, as the amount of traffic around a parcel, the amount of traffic at a particular time of year, for example, during a festival or other event.

(25) The parcel information may include a time slot in which other parking slots located within a predetermined range including a parcel are in use or the utilization rates for the parking slots. In other words, the parcel information may include a time slot in which vehicles are parked in other parking slots around a parcel. The information acquisition apparatus **40** may include a camera that images other parking slots and calculate, based on images of the parking slots, a time slot in which the parking slots are in use or the utilization rates for the parking slots. The information acquisition apparatus may include sensors installed in other parking slots and calculate, based on results of detecting that vehicles are parked in the parking slots, a time slot in which the parking slots are in use or the utilization rates for the parking slots. The information acquisition apparatus **40** may include an interface to acquire a time slot in which other parking slots are in use or the utilization rates for the parking slots from an external apparatus such as a parking management server that manages the parking slots. The information acquisition apparatus **40** may include an external apparatus such as a parking management server.

(26) The parcel information may include information regarding market parking fee rates for other parking slots located within a predetermined range including a parcel. The parcel information may include information on market parking fee rates for hourly parking slots and/or market parking fee rates for monthly parking slots. The information acquisition apparatus **40** may include a camera that images rate sheets displayed on other parking slots, and may acquire information regarding market parking fee rates for the parking slots based on the images of the rate sheets for the parking slots. The information acquisition apparatus **40** may include an interface to acquire information on market parking fee rates for other parking slots from an external apparatus such as a parking management server that manages the parking slots. The information acquisition apparatus **40** may include an external apparatus such as a parking management server.

(27) The parcel information may include an occupancy rate for dwelling units within a predetermined range including a parcel or the number of years for which residents have lived. The occupancy rate for dwelling units or the number of years for which residents have lived is also referred to as resident data. The information acquisition apparatus **40** may include an interface to acquire the resident data for the dwelling units within the predetermined range including the parcel from an external apparatus such as a resident data server that manages the resident data. The information acquisition apparatus **40** may include an external apparatus such as a resident data server.

(28) The parcel information may include information regarding various facilities such as public, commercial, or medical facilities located within a predetermined range including a parcel. The information regarding the facilities is also referred to as facility data. The information acquisition apparatus **40** may include an interface to acquire the facility data from an external apparatus such as a facility data server that manages the facility data. The information acquisition apparatus **40** may include an external apparatus such as a facility data server.

(29) <Terminal Apparatus **20**>

(30) The information processing system **1** further includes a non-essential terminal apparatus **20**. The terminal apparatus **20** is communicably connected to the information processing apparatus **10** or the information acquisition apparatus **40** via the network **30**. The terminal apparatus **20** includes a controller **22** and an interface **24**.

(31) The controller **22** may be configured with one or more processors or one or more dedicated circuits. The controller **22** may be configured to be identical or similar to the controller **12** of the

information processing apparatus **10**. The controller **22** may be configured to include a memory. The memory may be configured to be identical or similar to the memory **14** of the information processing apparatus **10**.

(32) The interface **24** may be configured to include an input device, an output device, or a communication device. The interface **24** may be configured to be identical or similar to the interface **16** of the information processing apparatus **10**. The interface **24** may be wired or wirelessly communicably connected to the information processing apparatus **10** or the information acquisition apparatus **40**.

(33) The terminal apparatus **20** may be carried by a user or may be installed in a predetermined location. The terminal apparatus **20** may be configured as a mobile terminal, such as a smartphone or a tablet, or a Personal Computer (PC), such as a notebook PC or a tablet PC. The terminal apparatus **20** is not limited to the above examples and may include various devices. The number of terminal apparatuses **20** included in the information processing system **1** is not limited to one and may be two or more.

(34) <Output Apparatus **50**>

(35) The information processing system **1** further includes a non-essential output apparatus **50**. The output apparatus **50** is communicably connected to the information processing apparatus **10**, the information acquisition apparatus **40**, or the terminal apparatus **20** via the network **30**. The output apparatus **50** may be installed in a parcel that can be used as a parking slot.

(36) The output apparatus **50** may include, for example, a display apparatus. When a parcel is assigned as a parking slot, the display apparatus may display the word “empty” or similar text to indicate that the parcel is available for parking a vehicle. When a parcel is not assigned as a parking slot, the display apparatus may display the word “full” or similar text to indicate that the parcel is unavailable for parking a vehicle. The display apparatus may display the word “monthly” or similar text when a parcel is assigned as a monthly parking slot. The display apparatus may display the word “hourly” or similar text when a parcel is assigned as an hourly parking slot. The display apparatus may display a variety of other characters, not limited to those in the examples, and may also display symbols, graphics, or images.

(37) The output apparatus **50** may include, for example, an entry restriction apparatus. The entry restriction apparatus may include a gate or a flap board that restricts entry of a vehicle into a parcel. When a parcel is assigned as a parking slot, the entry restriction apparatus may open the gate or lower the flap board to allow a vehicle to enter the parcel. When a parcel is not assigned as a parking slot, the entry restriction apparatus may close the gate or raise the flap board to prevent a vehicle from entering the parcel. When a parcel is assigned as a monthly parking slot, the entry restriction apparatus may open the gate or lower the flap board at operation of a user who has contracted to park on a monthly basis. The entry restriction apparatus may be equipped with a sensor that detects a vehicle contracted to park on a monthly basis, and open the gate or lower the flap board when the vehicle is detected.

(38) (Example Operations of Information Processing System **1**)

(39) In the information processing system **1** according to the present embodiment, the controller **12** of the information processing apparatus **10** acquires parcel information from the information acquisition apparatus **40**. The controller **12** may extract, based on the parcel information, a parcel that can be used as a parking slot. The controller **12** may determine, based on the parcel information, whether the parcel can be used as a parking slot. The controller **12** may assign, as a parking slot, the parcel that can be used as a parking slot. Specific example operations of the controller **12** assigning a parcel as a parking slot will be described below.

(40) <Estimation of Vacant Status of Parcel>

(41) The controller **12** can assign a parcel as a parking slot when the parcel is vacant. The controller **12** may estimate a vacant status of the parcel based on the parcel information. The controller **12** may estimate a vacant status of the parcel based on information representing a track record of a

vacant status of the parcel. The controller **12** may estimate a vacant status of the parcel based on images of the parcel. The controller **12** may estimate a vacant status of the parcel based on a detection result of a sensor installed in the parcel.

(42) The controller **12** may estimate, as the vacant status of the parcel, the length of time for which the parcel is vacant. The controller **12** may estimate, as the vacant status of the parcel, a trend of timing when the parcel becomes vacant. The controller **12** may estimate, as the trend of timing when the parcel becomes vacant, for example, a time slot in which the parcel becomes vacant. The controller **12** may estimate, as the trend of timing when the parcel becomes vacant, for example, the day of the week when the parcel becomes vacant, or a specific time of the month, such as early or late in the month or at the beginning or end of the month, when the parcel becomes vacant.

(43) The controller **12** may determine, based on an estimation result of the vacant status of the parcel, whether the parcel can be used as a parking slot. The controller **12** may determine, based on an estimation result of the vacant status of the parcel, whether the parcel can be used as a monthly or hourly parking slot. The controller **12** may calculate, based on an estimation result of the vacant status of the parcel, a time period during which the parcel can be used as a parking slot, and determine a time period during which the parcel is assigned as a parking slot.

(44) When the parcel is estimated to be vacant at a specific time slot or time period, for example, the controller **12** may determine that the parcel can be used as a parking slot at the specific time slot or time period. When the parcel is estimated to be vacant for an extended period of time, the controller **12** may determine that the parcel can be used as a monthly parking slot. When the parcel is estimated to be vacant at a particular time slot, the controller **12** may determine that the parcel can be used as a monthly parking slot for the particular time slot only.

(45) <Calculation of Potential as Parking Slot>

(46) The controller **12** may calculate, based on information regarding a certain parcel, potential for assigning the certain parcel as a parking slot. The potential for assigning the certain parcel as a parking slot may include, for example, a predicted value of a utilization rate for the certain parcel as a parking slot. In other words, the potential for assigning the certain parcel to a parking slot may include a predicted value of a percentage of time a vehicle is parked when the certain parcel is used as a parking slot. The potential for assigning the certain parcel to a parking slot may include, for example, a projected value of revenue from assigning the certain parcel as a parking slot.

(47) The controller **12** may estimate a parking demand for the certain parcel based on the information regarding the certain parcel, in order to calculate the potential for assigning the certain parcel as a parking slot.

(48) For example, the controller **12** may estimate, as the parking demand, a utilization rate when the certain parcel is assigned as a parking slot, based on utilization rates for other parking slots around the certain parcel. For example, the controller **12** may estimate, as the parking demand for the certain parcel, a time slot in which a utilization rate is high when the certain parcel is assigned as a parking slot, based on time slots in which other parking slots around the certain parcel are in use.

(49) For example, the controller **12** may estimate, as the parking demand, a utilization rate when the certain parcel is assigned as a parking slot, based on the amount of traffic on roads around the certain parcel. For example, the controller **12** may estimate, as the parking demand for the certain parcel, a time slot in which a utilization rate is high when the certain parcel is assigned as a parking slot, based on the amount of traffic on roads around the certain parcel for each time slot.

(50) For example, the controller **12** may estimate a demand when the certain parcel is assigned as a monthly parking slot, based on resident data around the certain parcel. The controller **12** may estimate how often residents are replaced (moved) based on the number of years for which the residents have lived. When there is a high turnover of residents, the controller **12** may estimate that there is a high demand for the certain parcel assigned as a monthly parking slot. When there is a low turnover of residents, the controller **12** may estimate that there is a high demand for the certain

parcel assigned as a monthly parking slot. The controller **12** may estimate a time period in which other parking slots around the certain parcel become available based on the number of years for which the residents have lived.

(51) For example, the controller **12** may estimate, as the parking demand, a utilization rate when the certain parcel is assigned as a parking slot, based on facility data around the certain parcel. For example, the controller **12** may estimate, as the parking demand, a time slot in which a utilization rate is high when the certain parcel is assigned as a parking slot, based on time slots in which facilities around the certain parcel are heavily used. For example, the controller **12** may estimate a demand when the certain parcel is assigned as a monthly parking slot, based on parking space data on the facilities around the certain parcel. When there are few parking slots in the nearby facilities, the controller **12** may estimate that there is a large demand for monthly parking slots from the facilities. When there are few parking slots in the nearby facilities, the controller **12** may estimate that there is a large demand for hourly parking slots. When there is a temporary increase in the number of users of the nearby facilities, the controller **12** may estimate that there is a large demand for hourly parking slots.

(52) Based on the estimated parking demand for the certain parcel, the controller **12** may calculate, as the potential, a predicted value of a utilization rate for or revenue from assigning the certain parcel as a parking slot. The controller **12** may calculate each of potential for assigning the certain parcel to a monthly parking slot and potential for assigning the certain parcel to an hourly parking slot.

(53) The controller **12** may estimate a parking fee rate that can be set when the certain parcel is assigned as a parking slot, based on market parking fee rates for other parking slots around the certain parcel. Based on an estimated value of the settable parking fee rate, the controller **12** may calculate, as the potential, a predicted value of a utilization rate for or revenue from assigning the certain parcel as a parking slot. Based on an estimated value of the settable parking fee rate for an hourly parking slot, the controller **12** may calculate, as the potential, a predicted value of a utilization rate for or revenue from assigning the certain parcel as an hourly parking slot. Based on an estimated value of the settable parking fee rate for a monthly parking slot, the controller **12** may calculate, as the potential, a predicted value of a utilization rate for or revenue from assigning the certain parcel as a monthly parking slot.

(54) <Assignment as Parking Slot>

(55) The controller **12** may generate information suggesting to a lending user, such as an owner or manager of a parcel that is determined to be usable as a parking slot, that the parcel is assigned as either an hourly or monthly parking slot. The controller **12** may generate information suggesting to a lending user, such as an owner or manager of a parcel that is determined to be usable as a parking slot, that the parcel can be assigned as either an hourly or monthly parking slot. The information suggesting to assign the parcel as a parking slot is also referred to as assignment suggestion information. The controller **12** may output the assignment suggestion information to the terminal apparatus **20** of the lending user.

(56) The controller **12** may output, as the assignment suggestion information, a calculation result of potential for assigning the parcel as a parking slot. When the parcel is usable as either an hourly or monthly parking slot, the controller **12** may output, as the assignment suggestion information, information allowing a comparison between a calculation result of potential for assigning the parcel as an hourly parking slot and a calculation result of potential for assigning the parcel as a monthly parking slot. The controller **12** may output, as the assignment suggestion information, a utilization prediction when the parcel is assigned as a parking slot. The controller **12** may output, as the assignment suggestion information, information regarding a parking fee rate that is set when the parcel is assigned as a parking slot. The controller **12** may output, as the assignment suggestion information, information suggesting that the parcel is assigned as an hourly parking slot only during specific daytime hours and as a monthly parking slot during the other time slot.

(57) The controller **22** of the terminal apparatus **20** may output the assignment suggestion information using the output device of the interface **24**, to notify the lending user of the assignment suggestion information. The controller **22** may accept, using the input device of the interface **24**, an input to determine whether the lending user accepts the suggestion to assign the parcel as a parking slot. The controller **22** may accept, using the input device of the interface **24**, an input to determine whether the lending user assigns the parcel as an hourly parking slot or a monthly parking slot. The controller **22** may accept, using the input device of the interface **24**, an input to determine a parking fee rate that is set when the lending user assigns the parcel as a parking slot. The controller **22** may output the inputs of the lending user to the information processing apparatus **10**.

(58) The controller **12** of the information processing apparatus **10** may determine to assign the parcel as a parking slot based on the input from the lending user. The controller **12** may determine, based on the input of the lending user, whether the parcel is assigned as an hourly parking slot or a monthly parking slot. The controller **12** may generate information representing that the parcel has been assigned as an hourly parking slot or a monthly parking slot. The information representing that the parcel has been assigned as a parking slot is also referred to as assignment determination information. The controller **12** may determine a parking fee rate to be set for the parking slot. The assignment determination information may include information on the parking fee rate to be set for the parking slot. The controller **12** may output the assignment determination information to the terminal apparatus **20** of the lending user. The controller **22** of the terminal apparatus **20** may output the assignment determination information using the output device of the interface **24**, to notify the lending user of the assignment determination information.

(59) The controller **12** may notify the assignment suggestion information to the terminal apparatus **20** that is in the possession of a representative of a contractor managing parking slots. Based on the assignment suggestion information, the representative may negotiate with an owner or manager of a parcel that can be used as a parking slot and enter into an agreement to assign the parcel as a parking slot.

(60) The lending user may wish to delegate, to the information processing apparatus **10**, determination whether to assign the parcel as a parking slot, determination whether to assign the parcel as an hourly or monthly parking slot, or determination of a parking fee rate to be set for the parking slot. The controller **22** of the terminal apparatus **20** may accept, by the input device of the interface **24**, and output, to the information processing apparatus **10**, an input indicating a willingness of the lending user to delegate to the information processing apparatus **10** a determination regarding assignment of the parcel as a parking slot. Upon being delegated by the lending user to make a determination regarding assignment of the parcel to a parking slot, the controller **12** of the information processing apparatus **10** may automatically determine whether to actually assign the parcel, which has been determined to be usable as a parking slot, as an hourly or monthly parking slot without generating the assignment suggestion information. The controller **12** may automatically determine a parking fee rate to be set for the parking slot. Upon determining to assign the parcel as a parking slot, the controller **12** may generate assignment determination information and output the assignment determination information to the terminal apparatus **20** of the lending user.

(61) The controller **12** may determine whether to actually assign the parcel as a parking slot, based on a calculation result of potential when the parcel is assigned as a parking slot. For example, the controller **12** may determine to assign the parcel as a parking slot when a predicted value of revenue from assigning the parcel as a parking slot is above a revenue threshold. For example, the controller **12** may determine to assign the parcel as a parking slot when a predicted value of a utilization rate when the parcel is assigned as a parking slot is above a utilization rate threshold. The revenue threshold or utilization rate threshold can be determined based on, for example, cost or the like incurred when the parcel is assigned as a parking slot. In other words, the controller **12** may determine to assign the parcel to a parking slot when the calculation result of the potential meet

conditions.

(62) Both the assignment suggestion information and the assignment determination information are information regarding assignment of the parcel as a parking slot, and also collectively referred to as assignment information.

(63) <Management of Parking Slot>

(64) The controller **12** of the information processing apparatus **10** may output the assignment determination information to a parking slot management server that manages parking slots. The parking slot management server may generate, based on the assignment determination information, a database to manage the parking slots. The parking slot management server may provide, based on the database, a service to introduce the parking slots to using users. This facilitates increasing utilization rates or revenue of the parking slots. The controller **12** may output the assignment determination information to the terminal apparatus **20** of the using user through an external apparatus such as the parking slot management server, or to the terminal apparatus **20** of the using user without passing through the external apparatus. The controller **12** may perform the functions of the parking slot management server. The parking slot management server may provide a service to introduce both hourly and monthly parking slots. The parking slot management server may provide a service to introduce hourly parking slots and/or monthly parking slots. The parking slot management server may provide a service to introduce parking slots available for either on a hourly or monthly basis. Introducing monthly parking slots makes it easier for residents, employees, or the like around the parcel to find a place to park for an extended period of time. In addition, the need for a garage certificate at the time of purchase of a vehicle could be met.

(65) The controller **12** may generate a utilization prediction for the parking slots and output the utilization prediction to the terminal apparatus **20** of the using user. This enables the using users to recognize vacant statuses of the parking slots, thus facilitating using the parking slots. In addition, the using users are more likely to visit the vicinity of the parking slots with vehicles when the using users have the prospect of being able to park the vehicles. The controller **12** may change some of the hourly parking slots to monthly parking slots, based on utilization predictions for the hourly parking slots. When lending users have not delegated determination of assignment of parcels as parking slots, the controller **12** may generate information suggesting changing the hourly parking slots to monthly parking slots, and notify the lending users of the information to acquire determination of the lending users.

(66) When a certain parcel is assigned as a parking slot, the controller **12** may output assignment determination information to the output apparatus **50** installed in the certain parcel. The output apparatus **50** may output information based on the assignment determination information so that using users can recognize a status of the certain parcel. For example, a display apparatus, as the output apparatus **50**, may indicate whether the certain parcel is assigned as a parking slot. For example, an entry restriction apparatus, as the output apparatus **50**, may open a gate or lower a flap board when the certain parcel is available for parking.

(67) <Example Procedure for Information Processing Method>

(68) The controller **12** of the information processing apparatus **10** according to the present embodiment may perform an information processing method including a procedure of the flowchart in FIG. 3, for example, to assign a parcel as a parking slot. In the procedure of the flowchart in FIG. 3, the controller **12** determines to assign a parcel as a parking slot. The information processing method may be implemented as an information processing program to be executed by the controller **12**. The information processing program may be stored on a non-transitory computer readable medium.

(69) The controller **12** acquires parcel information (step S1). Based on the parcel information, the controller **12** estimates a vacant status of a parcel (step S2). Based on the parcel information, the controller **12** estimates a parking demand around the parcel (step S3). Based on an estimation result of the parking demand, the controller **12** calculates potential of the parcel as a parking slot

(potential when the parcel is assigned as a parking slot) (step S4).

(70) The controller **12** determines whether the potential meets conditions (step S5). When the potential does not meet the conditions (NO in step S5), the controller **12** ends execution of the procedure of the flow chart in FIG. 3, without assigning the parcel as a parking slot. When the potential meets the conditions (YES in step S5), the controller **12** assigns the parcel as a parking slot (step S6). The controller **12** generates assignment determination information, as assignment information, and outputs the assignment determination information to the terminal apparatus **20** of a lending user and the like. (step S7). After performing the process of step S7, the controller **12** ends the execution of the procedure of the flowchart in FIG. 3.

(71) The controller **12** may, for example, perform an information processing method including the procedure of the flowchart in FIG. 4. In the procedure of the flowchart in FIG. 4, a lending user determines to assign a parcel as a parking slot.

(72) The controller **12** acquires potential of a parcel as a parking slot (potential when a parcel is assigned as a parking slot), which is calculated in steps S1 to S4 of FIG. 3 (step S11). The controller **12** generates assignment suggestion information (suggestion information) and outputs the assignment suggestion information to the terminal apparatus **20** of a lending user (step S12).

(73) The controller **12** determines whether there is an application by the lending user for assigning the parcel as a parking slot in the terminal apparatus **20** (step S13). Specifically, the controller **12** acquires, from the terminal apparatus **20**, and determines information input by the lending user in the terminal apparatus **20**. When there is no application for assigning the parcel as a parking slot (NO in step S13), the controller **12** ends the procedure of the flowchart in FIG. 4. When there is an application for assigning the parcel as a parking slot (YES in step S13), the controller **12** assigns the parcel as a parking slot (step S14). The controller **12** generates assignment determination information, as assignment information, and outputs the assignment determination information to the terminal apparatus **20** of the lending user and the like (step S15). After performing the process of step S15, the controller **12** ends the execution of the procedure of the flowchart in FIG. 4.

(74) <Summary>

(75) As described above, according to the information processing system **1**, information processing apparatus **10**, information processing method, and information processing program of the present embodiments, whether a parcel can be used as a parking slot can be determined. This facilitates effective use of vacant spaces usable for parking, as parking spaces. In addition, parcels can be assigned as hourly parking slots and/or monthly parking slots. In this way, parking slots can be provided in a form that meets demand by using users.

(76) While embodiments of the present disclosure have been described with reference to the drawings and examples, it is to be noted that various modifications and revisions may be implemented by those skilled in the art based on the present disclosure. Accordingly, such modifications and revisions are included within the scope of the present disclosure. For example, functions or the like included in each means, each step, or the like can be rearranged without logical inconsistency, and a plurality of means, steps, or the like can be combined into one or divided.

(77) Examples of some embodiments of the present disclosure are described below. However, it should be noted that the embodiments of the present disclosure are not limited to these.

(78) [Appendix 1] An information processing apparatus comprising:

(79) an interface configured to be able to acquire information regarding a parcel; and a controller configured to generate, based on the information regarding the parcel, information regarding assignment of the parcel as a parking slot.

[Appendix 2] The information processing apparatus according to appendix 1, wherein the controller is configured to: calculate, based on the information regarding the parcel, potential for assigning the parcel as a parking slot; and determine, based on a calculation result of the potential, whether to assign the parcel as a parking slot.

[Appendix 3] The information processing apparatus according to appendix 2, wherein the potential for assigning the parcel as a parking slot includes a predicted value of a utilization rate for the parcel as a parking slot, or a predicted value of revenue from assigning the parcel as a parking slot.

[Appendix 4] The information processing apparatus according to appendix 2 or 3, wherein the potential for assigning the parcel as a parking slot includes potential for assigning the parcel as a monthly parking slot, or potential for assigning the parcel as an hourly parking slot.

[Appendix 5] The information processing apparatus according to any one of appendices 1 to 4, wherein the controller is configured to: estimate a vacant status of the parcel based on the information regarding the parcel; and determine, based on an estimation result of the vacant status of the parcel, whether to assign the parcel as a parking slot.

[Appendix 6] The information processing apparatus according to appendix 5, wherein the controller is configured to determine, based on the estimation result of the vacant status of the parcel, whether to assign the parcel as a monthly parking slot or an hourly parking slot.

[Appendix 7] The information processing apparatus according to appendix 5 or 6, wherein the controller is configured to determine, based on the estimation result of the vacant status of the parcel, a time period during which the parcel is to be assigned as a parking slot.

[Appendix 8] The information processing apparatus according to any one of appendices 1 to 7, wherein the information regarding the parcel includes information representing a track record of a vacant status of the parcel.

[Appendix 9] The information processing apparatus according to appendix 8, wherein the information representing the track record of the vacant status of the parcel includes information representing a time slot in which the parcel is vacant, or a number or size of vehicles that can be parked in the parcel when the parcel is vacant.

[Appendix 10] The information processing apparatus according to appendix 8 or 9, wherein the information representing the track record of the vacant status of the parcel includes an image of the parcel.

[Appendix 11] The information processing apparatus according to appendix 10, wherein the image of the parcel includes images of the parcel at at least two different dates and/or times.

[Appendix 12] The information processing apparatus according to any one of appendices 1 to 11, wherein the information regarding the parcel includes an amount of traffic on a road passing through a predetermined range including the parcel, a time slot in which a parking slot located within the predetermined range including the parcel is in use or a utilization rate for the parking slot, or an occupancy rate for a dwelling unit in the predetermined range including the parcel or a number of years for which a resident has lived.

[Appendix 13] The information processing apparatus according to any one of appendices 1 to 12, wherein the controller is configured to output information assigning the parcel as a parking slot to a terminal apparatus carried by a user who lends or uses the parcel.

[Appendix 14] The information processing apparatus according to appendix 13, wherein the controller is configured to output, to the terminal apparatus, a utilization prediction when the parcel is assigned as a parking slot.

[Appendix 15] The information processing apparatus according to any one of appendices 1 to 14, wherein the controller is configured to output information assigning the parcel as a parking slot to an output apparatus installed in the parcel.

[Appendix 16] An information processing method comprising: acquiring information regarding a parcel; and generating, based on the information regarding the parcel, information regarding assignment of the parcel as a parking slot.

[Appendix 17] The information processing method according to appendix 16, further comprising: calculating, based on the information regarding the parcel, potential for assigning the parcel as a parking slot; and determining, based on a calculation result of the potential for assigning the parcel as a parking slot, whether to assign the parcel as a parking slot.

[Appendix 18] The information processing method according to appendix 16 or 17, further comprising: estimating a vacant status of the parcel based on the information regarding the parcel; and determining, based on an estimation result of the vacant status of the parcel, whether to assign the parcel as a parking slot.

[Appendix 19] The information processing method according to appendix 18, further comprising determining, based on the estimation result of the vacant status of the parcel, whether to assign the parcel as a monthly parking slot or an hourly parking slot.

[Appendix 20] An information processing program configured to cause a processor to execute operations, the operations comprising: acquiring information regarding a parcel; and generating, based on the information regarding the parcel, information regarding assignment of the parcel as a parking slot.

Claims

1. An information processing apparatus comprising: an interface configured to acquire information regarding a parcel, the information regarding the parcel including an occupancy rate for a dwelling unit within a predetermined range including the parcel or a number of years for which a resident has lived; and a controller configured to: estimate a parking demand for the parcel based on the information regarding the parcel, calculate, based on a result of the estimation, potential for assigning the parcel as a parking slot, determine, based on a result of the calculation, whether to assign the parcel as a parking slot, based on the determination of whether to assign the parcel as a parking slot, control an output apparatus installed in the parcel, wherein the output apparatus includes an entry restriction apparatus including a gate or a flap board, and the controller is further configured to based on determining to assign the parcel as a parking slot, control the entry restriction apparatus to open the gate or lower the flap board to allow a vehicle to enter the parcel, and based on determining not to assign the parcel as a parking slot, control the entry restriction apparatus to close the gate or raise the flap board to prevent a vehicle from entering the parcel.
2. The information processing apparatus according to claim 1, wherein the potential for assigning the parcel as a parking slot includes a predicted value of a utilization rate for the parcel as a parking slot, or a predicted value of revenue from assigning the parcel as a parking slot.
3. The information processing apparatus according to claim 1, wherein the potential for assigning the parcel as a parking slot includes potential for assigning the parcel as a monthly parking slot, or potential for assigning the parcel as an hourly parking slot.
4. The information processing apparatus according to claim 1, wherein the controller is configured to: estimate a vacant status of the parcel based on the information regarding the parcel; and determine, based on an estimation result of the vacant status of the parcel, whether to assign the parcel as a parking slot.
5. The information processing apparatus according to claim 4, wherein the controller is configured to determine, based on the estimation result of the vacant status of the parcel, whether to assign the parcel as a monthly parking slot or an hourly parking slot.
6. The information processing apparatus according to claim 4, wherein the controller is configured to determine, based on the estimation result of the vacant status of the parcel, a time period during which the parcel is to be assigned as a parking slot.
7. The information processing apparatus according to claim 1, wherein the information regarding the parcel includes information representing a track record of a vacant status of the parcel.
8. The information processing apparatus according to claim 7, wherein the information representing the track record of the vacant status of the parcel includes information representing a time slot in which the parcel is vacant, or a number or size of vehicles that can be parked in the parcel when the parcel is vacant.
9. The information processing apparatus according to claim 7, wherein the information representing

the track record of the vacant status of the parcel includes an image of the parcel.

10. The information processing apparatus according to claim 9, wherein the image of the parcel includes images of the parcel at at least two different dates and/or times.

11. The information processing apparatus according to claim 1, wherein the information regarding the parcel includes an amount of traffic on a road passing through the predetermined range including the parcel, a time slot in which a parking slot located within the predetermined range including the parcel is in use, or a utilization rate for the parking slot.

12. The information processing apparatus according to claim 1, wherein the controller is configured to output information assigning the parcel as a parking slot to a terminal apparatus carried by a user who lends or uses the parcel.

13. The information processing apparatus according to claim 12, wherein the controller is configured to output, to the terminal apparatus, a utilization prediction when the parcel is assigned as a parking slot.

14. The information processing apparatus according to claim 1, wherein the controller is configured to output information assigning the parcel as a parking slot to the output apparatus installed in the parcel.

15. An information processing method comprising: acquiring information regarding a parcel, the information regarding the parcel including an occupancy rate for a dwelling unit within a predetermined range including the parcel or a number of years for which a resident has lived; estimating a parking demand for the parcel based on the information regarding the parcel; calculating, based on a result of the estimation, potential for assigning the parcel as a parking slot; determining, based on a result of the calculation, whether to assign the parcel as a parking slot; based on the determination of whether to assign the parcel as a parking slot, controlling an output apparatus installed in the parcel, wherein the output apparatus includes an entry restriction apparatus including a gate or a flap board; based on determining to assign the parcel as a parking slot, controlling the entry restriction apparatus to open the gate or lower the flap board to allow a vehicle to enter the parcel; and based on determining not to assign the parcel as a parking slot, controlling the entry restriction apparatus to close the gate or raise the flap board to prevent a vehicle from entering the parcel.

16. The information processing method according to claim 15, further comprising: estimating a vacant status of the parcel based on the information regarding the parcel; and determining, based on an estimation result of the vacant status of the parcel, whether to assign the parcel as a parking slot.

17. The information processing method according to claim 16, further comprising determining, based on the estimation result of the vacant status of the parcel, whether to assign the parcel as a monthly parking slot or an hourly parking slot.

18. A non-transitory computer readable medium storing an information processing program configured to cause a processor to execute operations, the operations comprising: acquiring information regarding a parcel, the information regarding the parcel including an occupancy rate for a dwelling unit within a predetermined range including the parcel or a number of years for which a resident has lived; estimating a parking demand for the parcel based on the information regarding the parcel; calculating, based on a result of the estimation, potential for assigning the parcel as a parking slot; determining, based on a result of the calculation, whether to assign the parcel as a parking slot; based on the determination of whether to assign the parcel as a parking slot, controlling an output apparatus installed in the parcel, wherein the output apparatus includes an entry restriction apparatus including a gate or a flap board; based on determining to assign the parcel as a parking slot, controlling the entry restriction apparatus to open the gate or lower the flap board to allow a vehicle to enter the parcel; and based on determining not to assign the parcel as a parking slot, controlling the entry restriction apparatus to close the gate or raise the flap board to prevent a vehicle from entering the parcel.

19. The information processing apparatus according to claim 1, wherein the output apparatus

further includes a display, and the controller is further configured to based on determining to assign the parcel as a parking slot, control the display to output first information indicating that the parcel is available for parking a vehicle, and based on determining not to assign the parcel as a parking slot, control the display to output second information indicating that the parcel is unavailable for parking a vehicle.

20. The information processing apparatus according to claim 1, wherein the controller is further configured to estimate how often the resident is replaced, based on the number of years for which the resident has lived, to estimate the parking demand for the parcel.
